



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 12

Time : 3 Hrs. 20 Mins.

Answers

1. (3)	41. (2)	81. (4)	121. (3)	161. (2)
2. (2)	42. (2)	82. (2)	122. (4)	162. (4)
3. (4)	43. (2)	83. (2)	123. (2)	163. (1)
4. (2)	44. (3)	84. (2)	124. (4)	164. (3)
5. (3)	45. (1)	85. (1)	125. (4)	165. (4)
6. (3)	46. (1)	86. (2)	126. (1)	166. (2)
7. (3)	47. (3)	87. (2)	127. (2)	167. (2)
8. (1)	48. (2)	88. (4)	128. (1)	168. (4)
9. (1)	49. (1)	89. (3)	129. (4)	169. (4)
10. (4)	50. (2)	90. (4)	130. (2)	170. (2)
11. (1)	51. (1)	91. (2)	131. (2)	171. (2)
12. (4)	52. (3)	92. (2)	132. (1)	172. (4)
13. (3)	53. (4)	93. (4)	133. (2)	173. (2)
14. (3)	54. (4)	94. (3)	134. (4)	174. (2)
15. (4)	55. (2)	95. (1)	135. (3)	175. (4)
16. (3)	56. (4)	96. (3)	136. (1)	176. (4)
17. (3)	57. (2)	97. (2)	137. (3)	177. (1)
18. (1)	58. (4)	98. (3)	138. (4)	178. (2)
19. (4)	59. (1)	99. (1)	139. (2)	179. (3)
20. (4)	60. (3)	100. (4)	140. (2)	180. (4)
21. (3)	61. (1)	101. (3)	141. (2)	181. (1)
22. (3)	62. (1)	102. (1)	142. (4)	182. (2)
23. (4)	63. (3)	103. (3)	143. (3)	183. (1)
24. (4)	64. (4)	104. (4)	144. (3)	184. (2)
25. (2)	65. (4)	105. (4)	145. (4)	185. (2)
26. (3)	66. (1)	106. (2)	146. (1)	186. (2)
27. (3)	67. (3)	107. (4)	147. (4)	187. (2)
28. (1)	68. (2)	108. (2)	148. (1)	188. (3)
29. (4)	69. (3)	109. (2)	149. (4)	189. (2)
30. (2)	70. (4)	110. (3)	150. (3)	190. (4)
31. (4)	71. (1)	111. (3)	151. (3)	191. (2)
32. (4)	72. (3)	112. (4)	152. (3)	192. (4)
33. (3)	73. (2)	113. (1)	153. (4)	193. (3)
34. (3)	74. (3)	114. (2)	154. (3)	194. (1)
35. (3)	75. (3)	115. (4)	155. (4)	195. (3)
36. (1)	76. (1)	116. (3)	156. (3)	196. (2)
37. (4)	77. (4)	117. (2)	157. (2)	197. (3)
38. (1)	78. (4)	118. (2)	158. (4)	198. (2)
39. (2)	79. (1)	119. (4)	159. (1)	199. (4)
40. (1)	80. (2)	120. (1)	160. (3)	200. (3)



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Answers and Solutions

PHYSICS

SECTION - A

- Answer (3)
Slope of position-time gives velocity. Magnitude of slope is first decreasing then increasing.
- Answer (2)
 $H = \frac{1}{2}gt^2 \Rightarrow H = \frac{1}{2} \times 10 \times 8^2 \Rightarrow H = 320 \text{ m} \dots(i)$
and $t = \frac{40}{10} = 4 \text{ s}$
 $h' = \frac{1}{2}gt^2 \Rightarrow h' = \frac{1}{2} \times 10 \times 4^2 \Rightarrow h' = 80 \text{ m} \dots(ii)$
Height from ground = $H - h' = 320 - 80 = 240 \text{ m}$
- Answer (4)
Impulse $\Rightarrow \vec{I} = \Delta\vec{P} = \vec{P}_f - \vec{P}_i$
 $\vec{I} = m(\vec{v}_f - \vec{v}_i)$
 $|\vec{I}| = 3 \left(54 \times \frac{5}{18} - 36 \times \frac{5}{18} \right)$
 $|\vec{I}| = 3 \times 5 \times (3 - 2) = 15 \text{ kg m s}^{-1}$
- Answer (2)
From Pascal's law $\Rightarrow P_1 = P_2$
 $\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow \frac{10^6 \times 10^{-5} \text{ N}}{150 \text{ cm}^2} = \frac{F_2}{15 \times 10^5 \text{ cm}^2}$
 $F_2 = 10^5 \text{ N}$
- Answer (3)
We know, $U = nC_v T$
 $= \frac{5}{2} nRT = \frac{5}{2} (PV)$

- Answer (3)
Net electric flux is the measure of field lines originating/terminating from a charge.

$$\therefore \frac{N}{N'} = \frac{+3Q}{|-Q|} \Rightarrow N' = \frac{N}{3}$$

$$\text{And } N'' = N - \frac{N}{3} = \frac{2N}{3}$$

- Answer (3)
de-Broglie wavelength is given by

$$\lambda = \frac{h}{p}$$

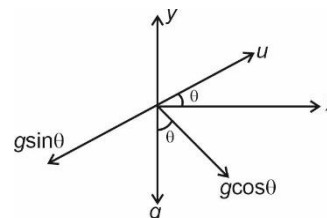
$\therefore$ If λ is same then p is same.

$$\text{Also, } KE = \frac{p^2}{2m} \Rightarrow KE \propto \frac{1}{m}$$

- Answer (1)

$$\left[\frac{A}{B} \right] = [Et] = [ML^2T^{-1}]$$

- Answer (1)



When velocity is perpendicular to initial velocity then

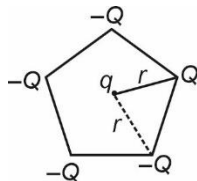
$$u - g \sin \theta t = 0$$

$$t = \frac{u}{g \sin \theta} = \frac{150 \times 2}{10 \times \sqrt{3}} = \frac{30}{\sqrt{3}} = 10\sqrt{3} \text{ s}$$

10. Answer (4)

Young's modulus depends upon material of wire of a wire.

11. Answer (1)



Let assume the charge (Q) is $Q + Q - Q$. Due to the $-Q$ charge at all corners net force is zero. So remaining $+2Q$ will give force $\frac{2kQq}{r^2}$.

12. Answer (4)

$$V_P - IR + E - \frac{Ldl}{dt} = V_Q$$

$$V_Q - V_P = -15 + 15 - 7 \times 10^{-3} \times 10^3 \times 2 = -14 \text{ V}$$

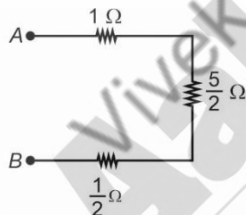
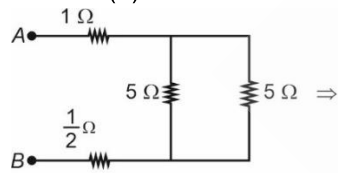
13. Answer (3)

Electric potential inside a conducting shell is $\frac{kQ}{R}$.

And electric potential outside it is given by

$$E = \frac{kQ}{r} \text{ for } r \geq R$$

14. Answer (3)



$$R_{eq} = 1 + \frac{5}{2} + \frac{1}{2} = 1 + \frac{6}{2} = 4 \Omega$$

15. Answer (4)

$$\text{Current sensitivity} \Rightarrow \frac{\theta}{i} = \frac{NBA}{C}$$

16. Answer (3)

Kinetic energy per molecule per degree of freedom is $\frac{1}{2}k_B T$

The rotational degree of freedom for diatomic gas is equal to 2

$$\therefore KE = \frac{2}{2}k_B T = k_B T$$

17. Answer (3)

$$\text{Wave number} = \frac{1}{\lambda}$$

18. Answer (1)

The magnetic force $q(\vec{v} \times \vec{B})$ will move positive charge towards end B.

19. Answer (4)

Net electric flux associated with the cube placed in a uniform electric field is zero.

$$\phi_{net} = \frac{q_{inside}}{\epsilon_0}$$

$$\Rightarrow q_{inside} = 0$$

20. Answer (4)

$$\text{Average power} = V_{rms} I_{rms} \cos \phi$$

$$= \frac{200}{\sqrt{2}} \times \frac{10}{\sqrt{2}} \times \cos\left(\frac{\pi}{3}\right)$$

$$= 1000 \times \frac{1}{2} = 500 \text{ W}$$

21. Answer (3)

Magnetic induction due to portion CD at O is zero

22. Answer (3)

$$PV^\gamma = \text{constant}$$

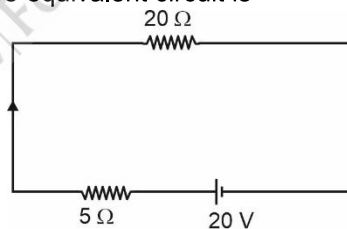
$$\& \text{ for an ideal gas, } PV = nRT$$

$$P = \frac{nRT}{V} \Rightarrow \frac{nRT}{V} V^\gamma = \text{constant}$$

$$TV^{\gamma-1} = \text{constant}$$

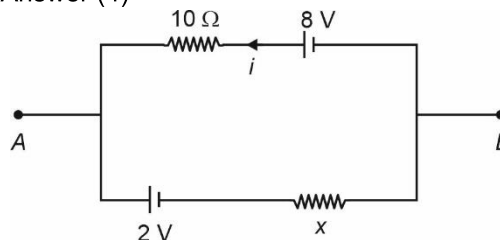
23. Answer (4)

Diode D_1 is in forward bias and D_2 is in reverse bias. So equivalent circuit is



$$I = \frac{20}{25} = \frac{4}{5} = 0.8 \text{ A}$$

24. Answer (4)



$$\Rightarrow V_A + 10i - 4 = V_B$$

$$\Rightarrow V_A - V_B - 4 = 10i$$

$$\Rightarrow 6 - 4 = 10i \Rightarrow i = \frac{2}{10} = \frac{1}{5} \text{ A}$$

$$\text{Now } V_A - 2 - xi = V_B$$

$$V_A - V_B - 2 = xi \Rightarrow 4 = \frac{x}{5} \Rightarrow x = 20 \Omega$$

25. Answer (2)

$$\text{Range of ammeter } I = I_g \left(1 + \frac{G}{S} \right)$$

$$\Rightarrow I = 3I_g \text{ (given)}$$

$$3I_g = I_g \left(1 + \frac{R}{S} \right)$$

$$\Rightarrow 3 = 1 + \frac{R}{S}$$

$$\Rightarrow \frac{R}{S} = 2$$

$$S = \frac{R}{2}$$

26. Answer (3)

By Newton's Law of cooling

$$\frac{T_1 - T_2}{t} = k \left[\frac{T_1 + T_2}{2} - T_s \right]$$

$$\Rightarrow \frac{10}{10} = k[65] \Rightarrow k = \frac{1}{65}$$

and

$$\frac{10}{t} = k[55] \Rightarrow \frac{10}{t} = \frac{1}{65} \times 55$$

$$t = \frac{65 \times 10}{55} = 11.8 \text{ minutes}$$

27. Answer (3)

0°C ice to 0°C water

$$\Rightarrow Q_1 = ml_f = 10 \times 80 = 800 \text{ cal}$$

0°C water into 80°C water

$$\Rightarrow Q_2 = ms\Delta T = 10 \times 1 \times 80 = 800 \text{ cal}$$

$$Q_{\text{required}} = 1600 \text{ cal}$$

28. Answer (1)

$$\lambda_m = \frac{\lambda_{\text{air}}}{\mu}$$

$$\mu = \frac{\lambda_{\text{air}}}{\lambda_m} = \frac{5000}{2500} = 2$$

By snell's law

$$\mu \sin i_c = 1 \cdot \sin 90^\circ$$

$$\sin i_c = \frac{1}{\mu}$$

$$i_c = \sin^{-1} \left(\frac{1}{2} \right) = 30^\circ$$

29. Answer (4)

$$P = 10D$$

$$\Rightarrow \frac{1}{f} = \frac{10}{100}$$

$$\Rightarrow f = 10 \text{ cm}$$

When image formed at infinity

$$\text{Magnifying power} = \frac{D}{f} = \frac{25}{10} = 2.5$$

30. Answer (2)

When light is incident at polarising angle, then reflected and refracted rays are perpendicular to each other.

$$\therefore \mu = \tan \theta_p$$

$$= \tan 53^\circ = \frac{4}{3}$$

31. Answer (4)

$$a = \frac{100 - 100 \times \sin 37^\circ}{20} = \frac{100 - 60}{20} = \frac{40}{20} = 2 \text{ m/s}^2$$

32. Answer (4)

$$\begin{array}{c} x=0 \qquad \qquad x=A/2 \\ \hline t=0 \end{array}$$

$$\Rightarrow x = A \sin(\omega t)$$

$$\Rightarrow \frac{A}{2} = A \sin(\omega t)$$

$$\Rightarrow \frac{\pi}{6} = \frac{2\pi}{T} t \Rightarrow t = \frac{T}{12}$$

33. Answer (3)

Moment of inertia about

$$\text{Given axis} = \frac{MR^2}{2}$$

$$\Rightarrow Mk^2 = \frac{MR^2}{2} \Rightarrow k = \frac{R}{\sqrt{2}}$$

$$= 5 \times 0.707 = 3.54 \text{ cm}$$

34. Answer (3)

$$\vec{R}_{\text{com}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3}{m_1 + m_2 + m_3}$$

$$= \frac{m(1, 2) + m(2, 3) + m(3, 4)}{3m} = \left(\frac{6}{3}, \frac{9}{3} \right)$$

$$= (2, 3)$$

35. Answer (3)

Time varying magnetic field produces electric field (Faradays experiment) & time varying electric field produces magnetic field. (Maxwells experiment)

SECTION-B

36. Answer (1)

Here magnetic field due to long straight wire is zero on z-axis.

37. Answer (4)

The diode enter into break down region.

$$I_s = \frac{10V - 5V}{500 \Omega} = \frac{1}{100} A = 10 \text{ mA}$$

$$I_{RL} = \frac{V_z}{R_L} = \frac{5V}{2 \text{ k}\Omega} = 2.5 \text{ mA}$$

$$I_z = 10 \text{ mA} - 2.5 \text{ mA} = 7.5 \text{ mA}$$

38. Answer (1)

$$PV^{-1} = \text{constant} \Rightarrow P = kV$$

$$W = \int PdV = k \left[\frac{V^2}{2} \right]$$

$$= \frac{1}{2} P_2 V_2 - \frac{1}{2} P_1 V_1$$

$$= \frac{R}{2} [T_2 - T_1] = \frac{RT_0}{2}$$

$$\Delta U = 1 \times \frac{3}{2} R \times T_0 = \frac{3}{2} RT_0$$

$$\text{Hence } Q = W + \Delta U = 2RT_0$$

39. Answer (2)

For two different light sources maximum intensity in the interference pattern is equal to $I_1 + I_2$

40. Answer (1)

$$V_{\text{net}} = \sqrt{V_R^2 + V_L^2}$$

$$10^2 = 5^2 + V_L^2 \Rightarrow V_L^2 = 100 - 25 = 75$$

$$V_L = 5\sqrt{3} \text{ V}$$

$$I = \frac{5\sqrt{3}}{5\sqrt{3}} A = 1 \text{ A}$$

41. Answer (2)

For minimum deviation condition

$$r = \frac{A}{2} = 30^\circ$$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 45^\circ}{\sin 30^\circ} = \sqrt{2}$$

42. Answer (2)

$$\Delta U = W_f$$

$$mgh = \mu mgx \Rightarrow \mu = \frac{h}{x} = \frac{1}{8}$$

43. Answer (2)

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \Rightarrow \frac{1}{+20} = (\mu - 1) \left(\frac{+1}{R} \right)$$

$$\frac{1}{20} = (\mu - 1) \times \frac{1}{40} \Rightarrow \mu - 1 = \frac{2}{3} \Rightarrow \mu = 1 + \frac{2}{3} = \frac{5}{3}$$

44. Answer (3)

According to de-Broglie wavelength $\lambda = \frac{h}{p}$ for all particles.

45. Answer (1)

$$V_{\text{max}} = A\omega \text{ \& } a_{\text{max}} = A\omega^2$$

$$\frac{a_{\text{max}}}{V_{\text{max}}} = \frac{A\omega^2}{A\omega} = \omega$$

$$\Rightarrow \frac{\pi}{2} = \omega = 2\pi f$$

$$f = \frac{1}{4} \text{ Hz}$$

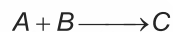
46. Answer (1)

$$\vec{F} = q(\vec{V} \times \vec{B})$$

Here magnetic field due to wire at point P is into the plane of the paper.

Therefore direction of force is opposite to OX

47. Answer (3)



For stable reaction

$$(B.E.)_P > (B.E.)_R$$

$$\Rightarrow E_c > E_a + E_b$$

48. Answer (2)

$$y = \vec{A} + \vec{B} = \vec{A} \cdot \vec{B}$$

NAND gate

49. Answer (1)

Transition I has highest energy of emission. So it has least wavelength

50. Answer (2)

$$g' = \frac{GM}{\left(\frac{R}{4} + R\right)^2} = \frac{16}{25} \frac{GM}{R^2} = \frac{16}{25} \times 10$$

$$= \frac{32}{5} = 6.4 \text{ m s}^{-2}$$

$$W = 640 \text{ N}$$

CHEMISTRY

SECTION - A

51. Answer (1)

(Quantum number)	(Orbital)
$n = 3, l = 2$	$3d$
$n = 3, l = 1$	$3p$
$n = 2, l = 1$	$2p$
$n = 4, l = 3$	$4f$

52. Answer (3)

Oxide	Nature
Cl_2O_7	Acidic
Al_2O_3	Amphoteric
Na_2O	Basic
N_2O	Neutral

53. Answer (4)

For first order reaction

$$t_{1/2} = \frac{0.693}{k} \Rightarrow k = \frac{0.693}{t_{1/2}}$$

$$k.t = 2.303 \log \frac{[A]_0}{[A]_t}$$

$$\frac{0.693}{t_{1/2}} \times (3.32) = 2.303 \log \frac{100}{70}$$

$$t_{1/2} = 6.45 \text{ second}$$

54. Answer (4)

They are chemically inert

55. Answer (2)



$$\Delta_r H^\circ_{\text{reaction}} = (\Delta_r H^\circ)_{\text{products}} - (\Delta_r H^\circ)_{\text{reactant}}$$

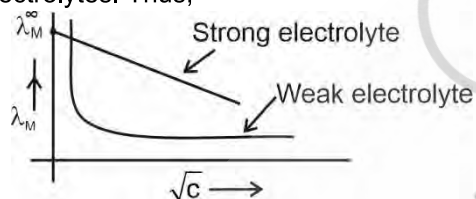
$$= (-650 - 500) - (-1300)$$

$$= -1150 + 1300$$

$$= +150 \text{ kJ/mol}$$

56. Answer (4)

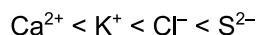
According to Ostwald's law of dilution, the degree of dissociation increases with the dilution of weak electrolytes. Thus,



57. Answer (2)

For isoelectronic species, the ionic size decreases with increase in nuclear charge.

∴ Order of ionic size is



58. Answer (4)

$$r_n = \frac{52.9(n^2)}{z} \text{ pm}$$

$$\frac{r_{\text{He}^+}}{r_{\text{Li}^{2+}}} = \frac{(3)^2}{2}$$

$$\frac{r_{\text{Li}^{2+}}}{3} = \frac{(3)^2}{2}$$

$$\frac{238.05}{r_{\text{Li}^{2+}}} = \frac{9}{2} \times \frac{3}{9}$$

$$\frac{238.05}{r_{\text{Li}^{2+}}} = \frac{3}{2}$$

$$r_{\text{Li}^{2+}} = \frac{238.05 \times 2}{3} = 158.7 \text{ pm}$$

59. Answer (1)

(Molecules)	(Shape)
BrF ₅	Square pyramidal
PF ₅	Trigonal bipyramidal
SF ₄	See-saw
XeF ₄	Square planar

60. Answer (3)

Bond orders	Species
2.5	O ₂ ⁺
2	O ₂
1	O ₂ ²⁻

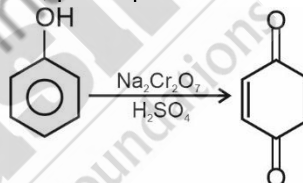
61. Answer (1)

Extensive property depends upon the quantity or size of matter present in the system.

62. Answer (1)

- Due to presence of -NO₂ group at para position, phenoxide ion gets stabilised.

So, p-nitrophenol is more acidic than phenol.



- In Kolbe's reaction phenol forms salicylic acid.

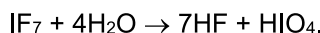
63. Answer (3)

Group V basic radicals precipitate as carbonates.

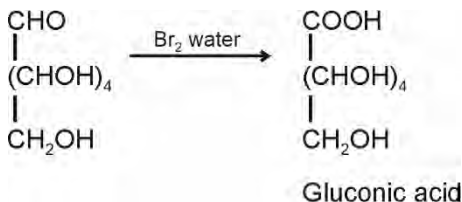
64. Answer (4)

F₂ does not show disproportionation reaction since fluorine does not show any positive oxidation state.

65. Answer (4)



66. Answer (1)



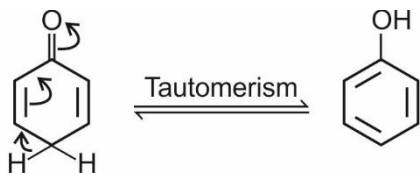
67. Answer (3)

Benzaldehyde gives positive Tollens test but does not give Fehling's test.

68. Answer (2)

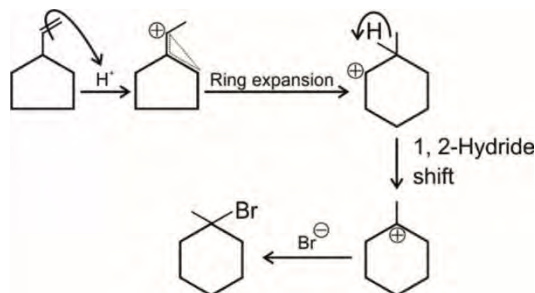
-NO₂ has -M effect, it decreases the stability of the carbocation.

69. Answer (3)

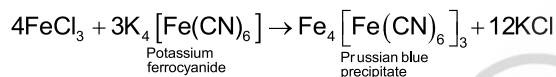


- enolisable H is present

70. Answer (4)



71. Answer (1)



72. Answer (3)

188 g AgBr contains 80 g of bromine

0.2 g of AgBr contains $\frac{80 \times 0.2}{188}$ g of bromine

$$\text{Percentage of bromine} = \frac{80 \times 0.2 \times 100}{188 \times 0.28}$$

= 30.4%

73. Answer (2)

$$\therefore \pi_{1(\text{urea})} = \pi_{2(\text{unknown})}$$

$$\therefore C_1 = C_2$$

$$\frac{W_1}{M_1} = \frac{W_2}{M_2}$$

$$\frac{5}{60} = \frac{2}{M_2}$$

$$M_2 = \frac{60 \times 2}{5} = 24 \text{ g/mol}$$

74. Answer (3)

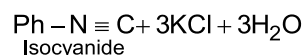
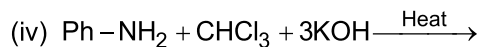
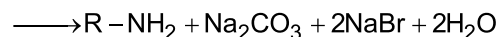
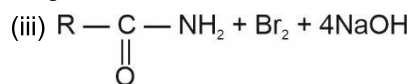
1 molal aq. solution means 1 mol of solute is dissolved in 1000 g of H₂O.

$$\text{Number of moles of H}_2\text{O} = \frac{1000}{18} = 55.55 \text{ mol}$$

$$\chi_{\text{solvent}} = \frac{55.55}{55.55 + 1} = 0.9823$$

75. Answer (3)

(i) The process of cleavage of the C-X bond by ammonia molecule is known as ammonolysis.

(ii) The reaction of phthalimide with KOH and R-X gives 1° amines as R-NH₂.

76. Answer (1)

Constituents of bronze : Cu + Sn

77. Answer (4)

Down the group, bond enthalpy for the dissociation of H-E bond decreases as a result, the acidic character increases.

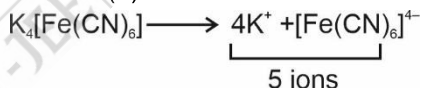
78. Answer (4)

Entropy for a spontaneous process increases till it reaches maximum and the change in entropy, $\Delta S = 0$.

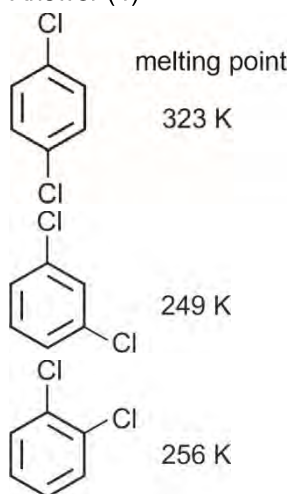
79. Answer (1)

For 10 moles of I⁻, 2 moles of MnO₄⁻ is required.For 2 mol of I⁻, $\frac{2}{5}$ moles of MnO₄⁻ is required.

80. Answer (2)



81. Answer (4)

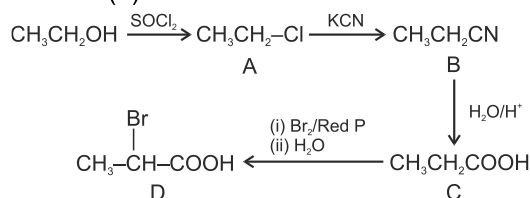


Para - isomer has high melting point than o and m isomer because of symmetry.

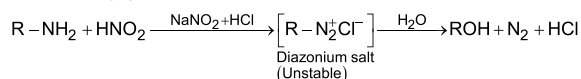
82. Answer (2)

The bond length of terminal B-H bond is less than the bond length of bridge B-H bond.

83. Answer (2)



84. Answer (2)

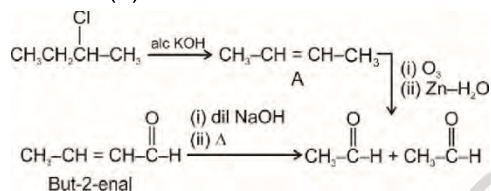


85. Answer (1)

Order of acidic character is
 $\text{HCOOH} > \text{C}_6\text{H}_5\text{COOH} > \text{C}_6\text{H}_5\text{CH}_2\text{COOH} > \text{CH}_3\text{COOH}$

SECTION-B

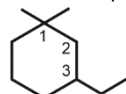
86. Answer (2)



87. Answer (2)

- Propene with $\text{HBr}/(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2$ = Free radical addition reaction.
- Methane with $\text{Br}_2/h\nu$ = Free radical substitution reaction.
- Propyne with HgSO_4 and dil. H_2SO_4 at 333 K = Electrophilic addition reaction.
- Benzene + $\text{CH}_3\text{Cl} \xrightarrow{\text{anhyd. AlCl}_3}$ electrophilic substitution reaction.

88. Answer (4)



3-Ethyl-1, 1-dimethylcyclohexane

89. Answer (3)

More stable the carbocation is, more will be the reactivity of the alkyl halide towards $\text{S}_{\text{N}}1$ reaction.

90. Answer (4)

In case of negative deviations from Rault's law, the intermolecular attractive forces between A-A and B-B are weaker than those between A-B and leads to decrease in vapour pressure resulting in negative deviation.

91. Answer (2)

A 38% solution of sulphuric acid is used as an electrolyte in lead storage battery.

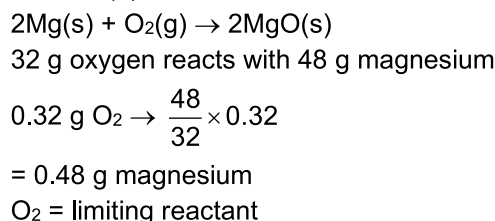
92. Answer (2)

Any acid stronger than carbonic acid gives effervescence of CO_2 with NaHCO_3 .

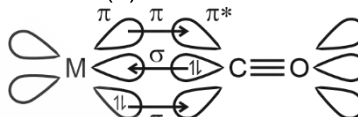
93. Answer (4)

As the forward reaction is exothermic and leads to lowering of number of gaseous molecule. Therefore, according to Le Chatelier's principle, upon increasing pressure and decreasing temperature, reaction proceeds forward.

94. Answer (3)



95. Answer (1)



Synergic bonding.

96. Answer (3)

$$\begin{aligned}
 \Delta H_{\text{rxn}} &= (E_a)_f - (E_a)_b \\
 -4.6 &= 9.6 - (E_a)_b \\
 (E_a)_b &= 9.6 + 4.6 \\
 &= 14.2 \text{ kJ mol}^{-1} \\
 \text{Also } (E_a)_b &> (E_a)_f \\
 [\Delta H_{\text{rxn}} &= \text{Negative}] \\
 \therefore \text{Reaction} &= \text{Exothermic}
 \end{aligned}$$

97. Answer (2)

Deficiency of vitamin B_{12} causes pernicious anaemia.

98. Answer (3)

NH_4OH in presence of NH_4Cl is the group reagent for Al^{3+} .

99. Answer (1)

$$\text{CaF}_2 \rightleftharpoons \underset{(S)}{\text{Ca}^{2+}} + \underset{(2S)^2}{2\text{F}^-}$$

$$K_{\text{sp}} = 5.3 \times 10^{-11} = (S)(2S)^2$$

In presence of 0.02 M NaF

$$\text{CaF}_2 \rightleftharpoons \underset{(S)}{\text{Ca}^{2+}} + \underset{(2S+0.02)^2}{2\text{F}^-}$$

$$(2S + 0.02) \approx (0.02)$$

$$K_{\text{sp}} = (S)(0.02)^2$$

$$\frac{5.3 \times 10^{-11}}{(0.0004)} = S$$

$$1.325 \times 10^{-7} = S$$

100. Answer (4)

PbS	–	Black ppt
$\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$	–	Violet colour
AgI	–	Dark yellow ppt
$[\text{Ce}(\text{NO}_3)_4(\text{ROH})_3]$	–	Red colour

BOTANY**SECTION - A**

101. Answer (3)

Mushroom belongs to class Basidiomycetes. In members of Basidiomycetes mycelium is branched and septate. The sex organs are absent. Sexual spores *i.e.*, basidiospores are exogenously produced on the basidium.

102. Answer (1)

Klinefelter's syndrome is caused due to the presence of an additional copy of X chromosome resulting into a karyotype of 47, XXY. Such an individual shows gynaecomastia or development of breasts.

103. Answer (3)

'Biodiversity hotspots' are characterised by a high degree of endemism. These hotspots are also regions of accelerated habitat loss.

104. Answer (4)

Eukaryotic transcription is a template dependent process which leads to the formation of hnRNA.

105. Answer (4)

False fruit develops from other floral parts and thalamus along with the development of ovary wall.

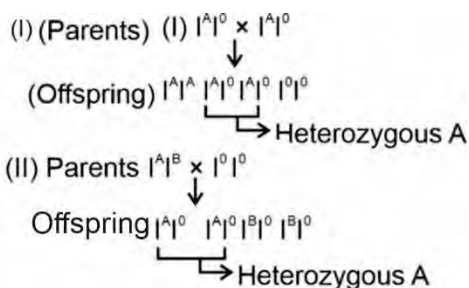
106. Answer (2)

Cyclosporin A is produced by the fungus *Trichoderma polysporum*.

107. Answer (4)

Heterozygous alleles for blood group A = $I^A i^0$

If Parents have genotype

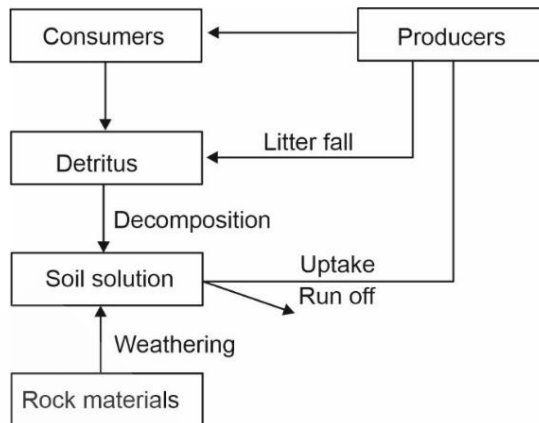


108. Answer (2)

Photorespiration results in the release of CO_2 with the utilisation of ATP. C_3 plants show photorespiration in which RuBisCO binds to O_2

109. Answer (2)

Phosphorus cycle is an example of sedimentary cycle.



110. Answer (3)

Ethylene initiates sprouting of potato tubers.

111. Answer (3)

In some prokaryotes like cyanobacteria, chromatophores contain photosynthetic pigments which are the membranous extensions into the cytoplasm.

112. Answer (4)

The beginning of diplotene is recognised by the dissolution of synaptonemal complex.

113. Answer (1)

Since, $2n = 18$

∴ Total number of bivalents

$$= \frac{1}{2} \text{ (total number of chromosomes)}$$

$$= \frac{1}{2} \times 18 = 9$$

114. Answer (2)

Members of kingdom Plantae exhibits only autotrophic mode of nutrition while, members of kingdom Fungi and Animalia exhibit only heterotrophic mode of nutrition.

115. Answer (4)

Some epidermal cells from the region of maturation form very fine and delicate thread-like structures called root hairs.

116. Answer (3)

In monocot root, there is no secondary growth because cambium is absent.

117. Answer (2)

PS II occurs on the inner surface of the thylakoids.

118. Answer (2)

Fructose-6-phosphate is phosphorylated to form fructose-1,6-bisphosphate. The enzyme which helps in transfer of phosphate from ATP to fructose-6-phosphate is phosphofructokinase (Pacemaker enzyme of EMP pathway).

119. Answer (4)

Phyllodes are green coloured leaf like petioles which perform photosynthesis. These are found in plants such as Australian acacia. In *Euphorbia* phylloclade is present.

120. Answer (1)

For every CO₂ molecule entering Calvin cycle, three molecules of ATP and two molecules of NADPH are required.

121. Answer (3)

In plants the tonoplast facilitate the transport of a number of ions and other materials against the concentration gradient into the vacuole. Hence, the concentration is significantly higher in the vacuole than in the cytoplasm.

122. Answer (4)

In racemose inflorescence, the shoot axis continue to grow indefinitely and the flowers are borne in an acropetal succession *i.e.* younger flowers are present towards the apex.

123. Answer (2)

Endodermis is absent in monocot stem.

124. Answer (4)

The flowers pollinated by insects or other animals have pollen grains which are sticky in nature.

125. Answer (4)

i gene of *lac* operon synthesises repressor protein.

The repressor protein binds to the operator region of the operon and prevent RNA polymerase from transcribing the operon.

126. Answer (1)

A typical mammalian cell contains 6.6×10^9 bp.

Since, a typical nucleosome contains 200 bp of DNA helix (2×10^2 bp) wrapped around histone octamer, the number of nucleosomes

$$= \frac{6.6 \times 10^9 \text{ bp}}{2 \times 10^2 \text{ bp}} = 3.3 \times 10^7$$

127. Answer (2)

Sacred groves/lakes found in several parts of India are

1. Khasi and Jaintia hills in Meghalaya
2. Aravalli hills of Rajasthan
3. Western Ghats regions of Karnataka and Maharashtra
4. Khecheopalri lake in Sikkim

128. Answer (1)

Geometric population growth equation is represented by $\frac{dN}{dt} = rN$

129. Answer (4)

Sickle cell anaemia is caused by substitution of glutamic acid by valine at the sixth position of the beta globin chain of the haemoglobin molecule. This is due to a transversion mutation.

130. Answer (2)

Meiosis involve pairing of homologous chromosomes and recombination between non-sister chromatids of homologous chromosomes. Four haploid cells are formed at the end of meiosis II.

131. Answer (2)

$(n - 1) \times (n - 1) \Rightarrow 2n - 2$ reflects nullisomic condition.

132. Answer (1)

Anthropogenic ecosystem does not possess self regulatory mechanism.

133. Answer (2)

There are 1300 species of birds in Amazonian rain forest.

134. Answer (4)

Altitude sickness can be experienced at high altitudes where body does not get enough oxygen due to low atmospheric pressure and causes nausea.

Altitude sickness symptoms includes nausea, fatigue and heart palpitations.

135. Answer (3)

In taxonomic hierarchy, the number of similar characters of organisms belonging to the particular category decreases from lowest to highest rank, *i.e.*, from species to kingdom.

136. Answer (1)

If the female parent produces unisexual flowers, there is no need for emasculation during artificial hybridisation.

137. Answer (3)

Lodicules are found in the members of family Poaceae.

138. Answer (4)

Joseph Priestley performed series of experiments that revealed essential role of air in the growth of green plants.

139. Answer (2)

Cuticle is present on aerial parts of plants.

140. Answer (2)

Humus is dark coloured amorphous substance.

141. Answer (2)

Repetitive sequence are stretches of DNA that are repeated many times. They are thought to have no direct coding function.

142. Answer (4)

During treatment of waste water, the primary treatment is a physical process which involves removal of large and small particles through filtration and sedimentation.

143. Answer (3)

Members of Phaeophyceae have mannitol and laminarin as stored food. They contain cellulose and algin in their cell wall.

Members of red algae contain cellulose, pectin and polysulphate esters in their cell wall.

144. Answer (3)

Eurythermal organisms include most of the mammals and birds.

145. Answer (4)

Funaria is a bryophyte. It exhibits haplo-diplontic life cycle pattern. In bryophytes the dominant phase is represented by haploid gametophyte and it alternates with multicellular sporophyte.

146. Answer (1)

Albugo is a parasitic fungi on mustard.

147. Answer (4)

Stem tendrils occur in most of the plants of Cucurbitaceae family such as, watermelon, cucumber, pumpkin.

148. Answer (1)

$$RrYy \times RrYy$$

$$\downarrow$$

$$RrYy \rightarrow 4$$

$$rrYy \rightarrow 2$$

$$Rryy \rightarrow 2$$

$$rryy \rightarrow 1$$

149. Answer (4)

Gas vacuoles are inclusion bodies in blue green algae.

150. Answer (3)

In meiosis I products, the chromosome number will be half and each chromosome has two chromatids, hence the total chromatids will be 24. The amount of DNA becomes double in S-phase and then reduced to half in meiosis I product.

ZOOLOGY

SECTION - A

151. Answer (3)

Compound epithelium is made of more than one layer of cells and thus has a limited role in secretion and absorption.

They cover dry surface of the skin and the moist surface of buccal cavity.

152. Answer (3)

When action potential is conducted to ventricular side by the AVN, the ventricular muscles contract (ventricular systole) and simultaneously atria undergo relaxation (atrial diastole) coinciding with the ventricular systole.

153. Answer (4)

All changes that occurred are the result of the activation of fight and flight response due to sympathetic stimulation. Sympathetic stimulation increases the cardiac output.

154. Answer (3)

Role of O_2 in the regulation of respiratory rhythm is quite insignificant.

Changes in CO_2 and H^+ concentration are mainly recognised for regulation of respiration.

155. Answer (4)

Selection of recombinants using pBR322 is a cumbersome procedure as it requires simultaneous plating on two plates having different antibiotics.

156. Answer (3)

Normal activities of the human heart are regulated intrinsically, i.e., autoregulated by specialised muscles present in heart (nodal tissues), hence it is called myogenic.

157. Answer (2)

Sternum, coccyx, atlas and axis are the bones of axial skeleton.

The bones of the limbs along with their girdles constitute the appendicular skeleton.

158. Answer (4)

Bt toxin gene has been introduced in plants in order to provide resistance against lepidopterans, dipterans and coleopterans.

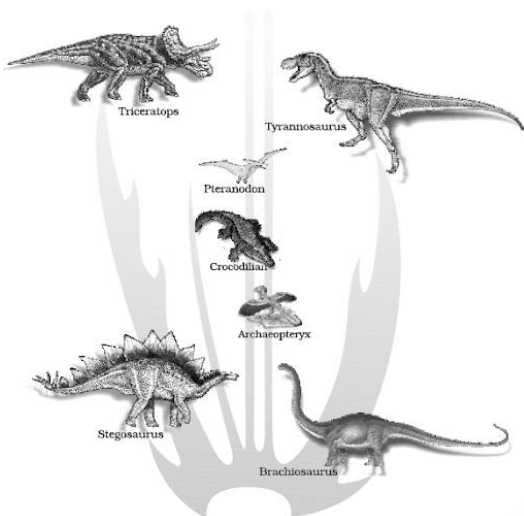
159. Answer (1)

Muscle fibre is the anatomical unit of muscle. Each muscle fibre has many parallelly arranged myofibrils. Each myofibril contains many serially arranged units called sarcomere which are the functional units.

160. Answer (3)

Tertiary structure of proteins is stabilised by peptide bonds, hydrogen bonds, ionic bonds, covalent bonds, hydrophobic bonds and Van der Waals interactions.

161. Answer (2)



162. Answer (4)

Anopheles, Aedes, Apis – Arthropoda

Sycon, Spongilla – Porifera

Canis – Chordata

163. Answer (1)

Reptiles, birds and land snails are uricotelic in nature. Mammals are ureotelic animals.

164. Answer (3)

Bone marked as 'X' is the sphenoid bone which forms the base of the cranium. It is not a facial bone.

165. Answer (4)

Both proximal convoluted tubule (PCT) and distal convoluted tubule (DCT) are present in renal cortex, outside the medullary pyramids.

166. Answer (2)

Ball and socket joint – Between humerus and pectoral girdle

Hinge joint – Between femur and tibia

Saddle joint – Between carpal and metacarpal of thumb

167. Answer (2)

Depolarisation occurs due to the opening of voltage gated Na^+ channels whereas repolarisation occurs due to opening of voltage gated K^+ channels and closure of voltage gated Na^+ channels.

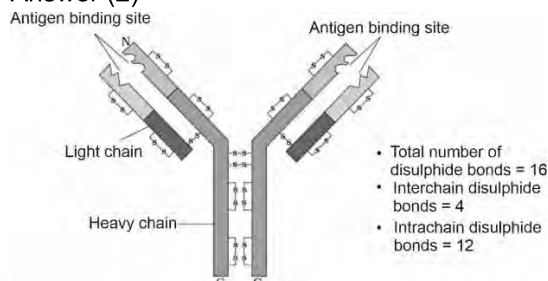
168. Answer (4)

GIFT involves *in vivo* fertilisation which means fertilisation takes place inside the female body.

169. Answer (4)

Aves are homeothermous, they have oil gland at the base of their tail, their hindlimbs have scales and are modified for walking, swimming whereas forelimbs are modified into wings. Endoskeleton is fully ossified (bony) and the long bones are hollow with air cavities (pneumatic).

170. Answer (2)



171. Answer (2)

Quaternary structure of proteins involves presence of more than one subunit of polypeptide chain as seen in insulin and in haemoglobin.

172. Answer (4)

'X' represents activation energy without an enzyme whereas 'Y' represents activation energy in presence of an enzyme. 'Z' represents transition state. If energy of product is at the lower level than substrate, the reaction is an exothermic reaction as one need not to supply energy (In the form of heat) in order to form the product.

173. Answer (2)

Reproductive system of male frog consists of a pair of yellowish ovoid testes. Vasa efferentia are 10-12 in number that arise from testes. They enter the kidney on their side and open into Bidder's canal. Finally, it communicates with the urinogenital duct that comes out of the kidneys and opens into the cloaca.

174. Answer (2)

Frogs respire on land and in water by two different methods. In water, skin acts as aquatic respiratory organ and on land the buccal cavity, skin and lungs act as the respiratory organs.

175. Answer (4)

Lane Q represents 'B' DNA fragments, R represents 'A' DNA fragments, and S represents 'C' DNA fragments.

176. Answer (4)

In cytokine barriers of innate immunity, interferons are included which protect non-infected cells from viral infections.

Also, cancerous patients are given α -interferons which activate their immune system and help in destroying the tumor.

177. Answer (1)

When foreign DNA (containing gene of interest) and vector DNA both are cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky-ends' and these can be joined together (end-to-end) using DNA ligases.

178. Answer (2)

In order to transform a bacterial cell, means to take an alien piece of DNA, the cells must first be made competent. This is done by treating them with a specific concentration of a divalent cation, such as calcium which increases the efficiency with which DNA enters the bacterial cells through pores in its cell wall.

179. Answer (3)

Vaccines containing attenuated pathogens or antigenic preparations provide artificial active immunity. As given from outside, it allows body to fight against the weakened pathogens, so that memory against the pathogen can be induced by immune system and later used when actual pathogen infects.

180. Answer (4)

Sterilisation procedure in the female is called tubectomy, where a small part of fallopian tube is removed or tied up through a small incision in the abdomen or through vagina. These techniques are highly effective but their reversibility is very poor. It blocks gamete transport and thereby prevents conception. Females who undergo tubectomy do menstruate.

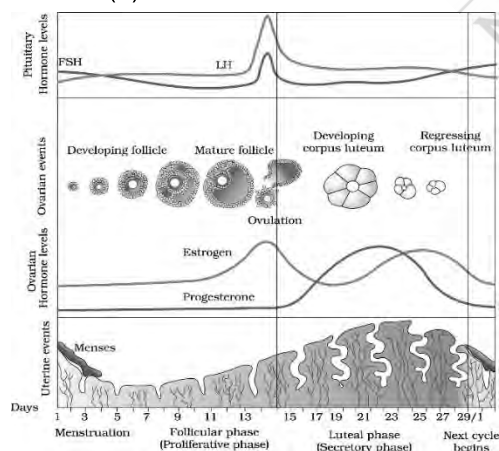
181. Answer (1)

Human sperm is a haploid cell, contain either 22 + X or 22 + Y chromosomes.

Primary spermatocyte is a diploid cell whereas secondary oocyte is a haploid cell.

Spermatids are non-motile as they do not have flagella.

182. Answer (2)



183. Answer (1)

The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull.

184. Answer (2)

DCT is not permeable to urea. Collecting duct allows passage of small amounts of urea into the medullary interstitium to keep up the osmolarity.

185. Answer (2)

The descending limb of loop of Henle is permeable to water but almost impermeable to electrolytes. This concentrates the filtrate as it moves downward whereas ascending limb of loop of Henle is impermeable to water but allows transport of electrolytes actively or passively which dilutes the filtrate as it moves upward.

SECTION - B

186. Answer (2)

AV node is known as the pace setter. It maintains the rhythm of the impulse conduction and thereby delay impulses by 0.1 second. It allows both the atria to empty completely.

187. Answer (2)

The cerebrum wraps around a structure called thalamus, which is a major coordinating centre for sensory and motor signalling.

188. Answer (3)

Diaphragm and external inter-costal muscles contract during inspiration and increase the volume of thoracic cavity thereby lowering the intra-pulmonary pressure than the atmospheric pressure.

189. Answer (2)

Fetal haemoglobin has higher affinity towards oxygen and shifts the curve in left from normal oxygen-dissociation curve. In the alveoli too, where there is high pO_2 , low pCO_2 , lesser H^+ concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin and curve is shifted towards left.

190. Answer (4)

The systemic circulation starts with the pumping of oxygenated blood by the left ventricle to the aorta which is carried to all the body tissues and the deoxygenated blood from there is collected by veins and returned to right atrium.

191. Answer (2)

All the mentioned features belong to organisms present in the phylum Echinodermata.

Sycon is a porifer, *Pila* is a mollusc.

Pleurobrachia is a ctenophore whereas *Antedon* is an echinoderm.

192. Answer (4)
During fertilization, a sperm comes in contact with the zona pellucida layer of the ovum and induces changes in the membrane that blocks the entry of additional sperms.
Zona pellucida disintegrates just before implantation.
193. Answer (3)
Urochordates and cephalochordates are acraniates which are primitive but referred to as invertebrate chordates or protochordates.
Cyclostomes are the most primitive of all craniates.
194. Answer (1)
Seminal vesicles store the sperms and glue them together to form spermatophore. A pair of spermatheca is present in the 6th abdominal segment in female cockroaches.
195. Answer (3)
The genetic code is nearly universal, for example, from bacteria to human UUU would code for phenylalanine.
Hence, genetic codes responsible for synthesis of polypeptides of insulin code the same proteins in both bacteria and humans.
196. Answer (2)
Myasthenia gravis is an autoimmune disorder affecting neuro-muscular junction leading to fatigue, weakening and paralysis of skeletal muscle. Osteoporosis is an age-related disorder.
- Sickle cell anaemia is a genetic disorder and can be detected by amniocentesis.
197. Answer (3)
PTH secreted by parathyroid gland is a hypercalcaemic hormone. Thymosins secreted by thymus, function in differentiation of T-lymphocytes. Glucagon secreted by α - cells of Islets of Langerhans in pancreas increases the blood glucose levels.
198. Answer (2)
Homology indicates common ancestry. It is based on divergent evolution, e.g., Thorns of *Bougainvillea* and tendrils of *Cucurbita*.
199. Answer (4)
Using *Agrobacterium* vector, nematode specific genes were introduced into the host plant. The introduction of DNA was such that it produced both sense and anti-sense RNA in the host cells. These two RNA's being complementary to each other form a double stranded RNA (dsRNA) that initiates RNAi, and this, silenced the specific mRNA of the nematode.
200. Answer (3)
Steroid hormones do not require the formation of second messengers. cAMP and IP₃ are the second messengers.

□ □ □